Nikhil Mehta

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RESEARCH INTERESTS

Deep Learning, Machine Learning, Probabilistic Models, Optimization

EDUCATION

Delhi Technological University, New Delhi, India — 2010-2014
B.Tech. in Computer Software Engineering

RESEARCH EXPERIENCE

Generative models and their applications — July '17 - Ongoing
Guided by Prof. Piyush Rai at IIT Kanpur

- Implemented an extension of the vanilla recurrent variational auto-encoder. The architecture had a combination of encoder-generator and a discriminator (trained in wake-sleep fashion) that allowed the network to learn structured disentangled representations. Discriminator performance for text classification improved by 3-4% with only 3000 labeled examples. [code]
- Implemented a wasserstein generative-adversarial-network (W-GAN) to synthesize data for unseen classes conditioned on class attributes. Preliminary results suggest a competitive zero-shot classification accuracy on Caltech-UCSD-Bird Dataset. [Ongoing] [code]
- Currently working on Bayesian non-parametric versions of deep learning models.

Learning based Localization of Autonomous Vehicles — Mar - Jun '17
Guided by Prof. G. Pandey at IIT Kanpur

- Created localization datasets (based on ground-reflectivity maps) from 3D lidar scans for machine learning tasks such as 1) Pose-Regression and 2) Metric Learning.
- Implemented a CNN based discriminative metric model to learn invariant representations which were used to calculate similarity scores between prior and local maps.
- Implemented a capsule based architecture to learn equivariant representations which could be used for pose regression. Open-sourced a proof of concept implementation of "Transforming Autoencoders" to learn transformations. [code]

MI based 3D Scan Registration for Autonomous Vehicles — Oct'16 - Feb '17
Guided by Prof. G. Pandey at IIT Kanpur

- Designed a fast Mutual Information based scan alignment algorithm for autonomous vehicles. Speed and robustness was found to be better than current state of the art scan matching methods when tested on two real time dynamic datasets (containing both rural and urban areas) recorded using a Lidar.
- Worked on a super-voxel clustering algorithm for 3D point clouds to represent an efficient Gaussian Mixture Model (GMM). This model was then used to apply a point-to-distribution alignment method for registration.

PUBLICATION

Nikhil Mehta, James R. McBride & Gaurav Pandey, **"Robust and Fast 3D Scan Alignment"** [Under Review at ICRA'18] [pdf]

INDUSTRY EXPERIENCE

Senior Project Associate at IIT-Kanpur — August 2016 - Current

- Research Assistant in the Machine Learning group at the Department of Computer Science. Working at the intersection of probabilistic machine learning and deep learning models.
- Research Assistant in the Autonomous Vehicle Group. Worked on applications of Deep Learning and Sensor Fusion for self-driving vehicles.

Software Engineer at Fitso Jan - Aug '16

Worked on a real time tracking and video application, allowing consumers to improve their general fitness. Was among the first 5 in the tech team. Our work here had an impact on more than 50,000 daily users seeking to improve their fitness. Application was given the Top Developer badge by Google after six months of launch.

Software Engineer at Zomato June '14 - Jan '16

Developed server side APIs and Android application for various consumer and enterprise products provided at Zomato across 22 countries including India, US and Canada. Developed a classification algorithm for restaurant reviews for spam filtering using neural network.

SELECTED PROJECTS & INTERNSHIPS

Research Intern at University of Birmingham, UK Dec '13 - Jan '14

Developed a paradigm and an experimental game in Python to investigate how people's perception of a hidden partner affect their behavior while competing against an artificial intelligent bot. Used the existing speech recognition model to create an end-end application for voice based commands.

Research Intern at University of Leeds, UK Jul - Aug '13

Designed a vision algorithm for cell (lamallepodia) tracking in DIC microscopic videos using entropy differences in subsequent frames. Real-time implementation was further improved by implementing the Extended Kalman filter.

Severity prediction using data mining and machine learning

Final year major project guided by Prof. Ruchika Malhotra at DTU Jan - May '14

Extracted multiple features from various open source dataset documentations using tf-idf, bag-of-words and word2vec. Formulated a multi-class classification problem using an ensemble of various models such as decision trees, support vector machines and naive bayes classifiers for severity prediction.

Hierarchical Reinforcement Learning for Line Follower

Robotics club at DTU Mar - Dec '13

Worked on the movement-planning layer for a line follower that had to find the optimal path to reach the goal position in a complex maze. Implementation allowed policy learning directly from the experience by decomposing the problem into a hierarchical sub-structure.

Mobile based university attendance

Minor project guided by Prof. Ruchika Malhotra at DTU Aug - Sep '12

Made an android application that allowed instructors to use their smartphone to mark attendance, publish announcements and set reminders for classes. All the data was synced to the web, providing a easy integration between the web and mobile application. This project was selected in the "Role of Technology in Saving the Environment" conference at DTU.

TECHNICAL SKILLS

Deep Learning Frameworks: Tensorflow, Pytorch
Languages: C, C++, Java, CUDA for GPU, Python, MATLAB, LaTeX.
Web Development: HTML5, CSS, JavaScript, PHP, SQL.
Proficient Algorithmic Skills

RELEVANT COURSEWORK

Computer Science: Data Structures, Algorithms, Artificial Intelligence, Compilers
Mathematics: Convex Optimization, Linear Algebra, Probabilistic Models
MOOCs: Machine Learning, Neural Networks for Machine Learning, Parallel GPU Programming using CUDA, Game Development [Certifications]